OJT Project Design Template

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Roll No(s):
Year & Section: 2025-2019 , BATCH-B
Project Title (as assigned): Role-Based Internship / OJT Management System
Project Type: Application Development
Stack / Framework:HTML5,CSS3,JavaScript (ES6+),React.js,Python,FastAPI,PostgreSQL, JWT,Chart.js.

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1. Problem Understanding

1.1 What is the problem statement in your own words?

Educational institutions and organizations face difficulties in managing OJT and internship programs due to manual task assignment, lack of real-time progress tracking, and unstructured evaluation methods.
Students, mentors, and administrators often rely on spreadsheets, emails, or verbal updates, which leads to inefficiency, miscommunication, and poor record maintenance.

The problem is the absence of a centralized, secure, and role-based system to manage internship workflows in a structured and transparent manner.

1.2 Why does this problem exist or matter?

This problem exists because:

- OJT programs involve multiple stakeholders (students, mentors, coordinators)
- There is no single platform to track tasks, submissions, and evaluations
- Manual processes are time-consuming and error-prone

Who benefits:

- Students: Clear tasks, deadlines, and feedback
- Mentors: Structured evaluation and progress tracking
- Institutions: Centralized data and improved monitoring

1.3 Key inputs and expected outputs:

Inputs	Process	Expected Outputs
User credentials	JWT authentication	Secure login
Task details	Task assignment logic	Assigned tasks
Submissions	Validation & storage	Submission records
Feedback	Evaluation processing	Progress updates
User actions	Role-based authorization	Controlled access

2. Functional Scope

2.1 What are the core features you plan to build (must-haves)?

2.1 Core features (Must-haves)

- Secure user authentication using JWT
- Role-Based Access Control (Admin, Mentor, Student)
- Task creation and assignment by mentors
- Task submission by students
- Feedback and evaluation system
- Progress tracking dashboard

2.2 What stretch goals could you attempt if time permits?

2.2 Stretch goals (If time permits)

- File upload support for submissions
- Email or in-app notifications
- Certificate generation after completion
- Analytics and performance reports
- Cloud deployment.
- Ranking based on performance in OJT.

2.3 Which libraries or tools will you use?

2.3 Libraries / Tools Used

- Frontend: React.js, HTML, CSS, JavaScript
- Backend: Python (FastAPI / Flask)
- Database: PostgreSQL / MySQL
- Authentication: JSON Web Token (JWT)
- Tools: Git, GitHub, Postman

3.System & Design Thinking

3.1 Sketch or describe your app flow / pipeline:

User Login
↓
JWT Authentication
↓
Role Verification
↓
Dashboard Access
↓
Task Assignment / Submission
↓
Feedback & Progress Update
↓
Data Stored in Database

3.1 Application flow / pipeline

3.2 What data structures or algorithms are central to this project?

3.2 Central data structures & logic

- Relational database tables (Users, Roles, Tasks, Submissions)
- Objects for API request/response handling.
- -Conditional logic for role-based permissions
- JWT token validation middleware

3.3 How will you test correctness or performance?

3.3 Testing strategy

- Manual testing of authentication and authorization
- API testing using Postman
- Role-based access testing
- Validation of task submission workflow
- UI testing for dashboards

4.Risks & Dependencies

4.1 What's the hardest part technically for you right now?

.1 Hardest technical challenges

- Implementing secure JWT authentication
- Designing role-based access correctly
- Managing relational database consistency
- Handling authorization edge cases

5.Timeline & Milestones (4 Weeks)

Week	Planned Deliverables	Mentor Checkpoint
W1	**Requirement analysis, problem understanding, finalizing features, PRD approval**	☐
W2	System architecture design, role definition (Admin, Mentor, Student), flow diagrams	☐
W3	Database schema design (Users, Roles, Tasks, Submissions), ER diagram	☐
W4	Backend project setup, environment configuration, basic API structure	☐
W5	User authentication module (Signup, Login, JWT implementation)	☐
W6	Role-Based Access Control (RBAC) implementation and protected routes	☐
W7	Admin module: user management, batch creation, mentor–student assignment	☐
W8	Mentor module: task creation, deadline setting, submission review	☐
W9	Student module: task viewing, submission upload, feedback viewing	☐
W10	Progress tracking dashboard, reports, basic analytics	☐
W11	Testing (API testing, role-based testing), bug fixing, UI improvements	☐
W12	Final documentation, demo preparation, PRD update, project submission	☐

5.2 What dependencies or help do you need from mentors?
5.2 Mentor support required

- Feedback on system design
- Guidance on best practices for security
- Review of database schema
- Suggestions for scalability

6.Evaluation Readiness

6.1 How will you prove that your project "works"?
6.1 Proof that the project works
Screenshots of dashboards/Video demo of task workflow/GitHub repository link/Live deployment (if applicable)/API testing screenshots

6.2 What success metric or goal will you aim for?
6.2 Success metrics

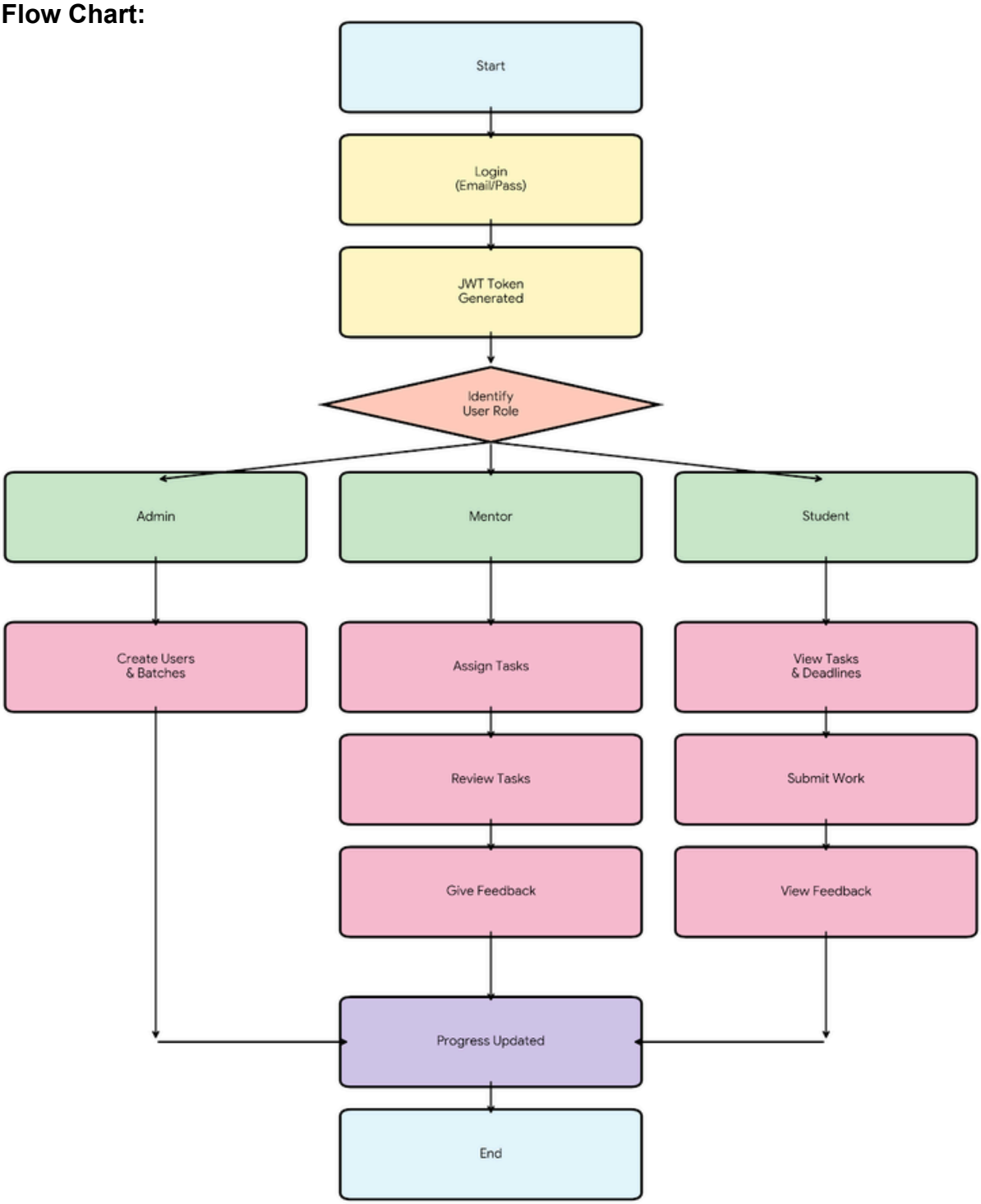
- 100% functional authentication & authorization
- All CRUD operations working
- Accurate progress tracking
- No unauthorized access between roles

7.Responsibilities

7.1 Responsibilities

Task	Student 1	Student 2	Notes
Authentication	◻	◻	JWT
Backend APIs	◻	◻	FastAPI
Frontend UI	◻	◻	React
Database Design	◻	◻	SQL
Documentation	◻	◻	PRD + Report

Flow Chart:



Signatures (Students):

Mentor Approval:

Date: